formulation on any special reference frame or any special class of reference frames whatsoever.[4] Sadly, the equations of differential geometry did not seem to admit such a law. Finally, in late winter, Einstein and Grossmann gave up the search and published the best law of warpage they could find—a law that relied for its definition on a special class of reference frames.

Einstein, eternally the optimist, managed to convince himself, briefly, that this was no catastrophe. To his physicist friend Paul Ehrenfest he wrote in early 1913, “What can be more beautiful than that this necessary specialization follows from [the mathematical equations for the conservation of energy and momentum]?” But after further thought he regarded it a disaster. He wrote to Lorentz in August 1913: “My faith in the reliability of the theory [the “law of warpage”] still fluctuates. . . . [Because of the failure to obey the generalized principle of relativity,] the theory contradicts its own starting point and all is up in the air.”

As Einstein and Grossmann struggled with spacetime curvature, other physicists scattered over the European continent took up the challenge of uniting the laws of gravity with special relativity. But none of them—Gunnar Nordstrom in Helsinki, Finland; Gustav Mie in Greifswald, Germany; Max Abraham in Milano, Italy—adopted Einstein’s spacetime curvature viewpoint. Instead they treated gravity, like electromagnetism, as due to a force field which lives in Minkowski’s flat, special relativistic spacetime. And no wonder they took this approach: The mathematics used by Einstein and Grossmann was horrendously complex, and it had produced a law of warpage that violated its authors’ own precepts.

Controversy swirled among the proponents of the various viewpoints. Wrote Abraham, “Someone who, like this author, has had to warn repeatedly against the siren song of [the principle of relativity] will greet with satisfaction the fact that its originator has now convinced himself of its untenability.” Wrote Einstein in reply, “In my opinion the situation does not indicate the failure of the relativity principle. . . . There is not the slightest ground to doubt its validity.” And privately he described Abraham’s theory of gravity as “a stately horse which lacks three legs.” Writing to friends in 1913 and 1914 Einstein said of the controversy, “I enjoy it that this affair is at least taken up with the requisite animation. I enjoy the controversies. Figaro mood: I’ll play him a tune.” “I enjoy it that colleagues occupy themselves at all with the theory [developed by Grossmann and me], although for the time being with the purpose of killing it. ... On the face of